Professor Messer Core 1 Companion Notes

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These notes follow the section order of Professor Messer's free 220-1201 video course. Watch a video, then use its table for a simple definition and an everyday example of each term. The explanations are my own words, not Professor Messer's notes.

Section 1: Mobile Devices

1.1 Laptop Hardware

Video: Laptop Hardware

Term	What it is	Example
Laptop battery	Stores power so the laptop works unplugged	A lunchbox of energy for when you're away from the wall
Keyboard / key caps	The keys can be popped off and replaced	Swapping a broken key like replacing one Lego brick
SSD upgrade	Replacing an old hard drive with a fast SSD	Trading a bicycle for a scooter; same trip, much faster
NFC	Wireless that only works when two things almost touch	Tapping your phone to pay at a store
Webcam and microphone	Built-in camera and mic, usually in the screen frame	Joining a video call for class

1.2 Mobile Device Configuration

Videos: Connecting Mobile Devices · Mobile Device Accessories

Term	What it is	Example
USB / USB-C	Cables that charge and move data	One USB-C cable charging your phone and laptop
Lightning	Apple's older charging port	The cable for an older iPhone

Term	What it is	Example
Bluetooth	Short-range wireless; devices must pair first	Two walkie-talkies agreeing on a secret channel
Hotspot / tethering	Sharing your phone's internet with other devices	Your phone turns into a mini Wi-Fi router for your laptop
Stylus	A pen for writing on a touchscreen	Drawing on a tablet like paper
Docking station	One plug that connects a laptop to monitors, keyboard and more	Plugging into a desk "home base"
Port replicator	A simpler dock that just adds extra ports	A power strip, but for USB and video ports
Trackpad / drawing pad	Touch surfaces for moving a cursor or drawing	Using your finger as the mouse

1.3 Mobile Connectivity

Videos: Mobile Device Networks · Mobile Device Management

Term	What it is	Example
Cellular (3G/4G/5G)	Internet through phone towers; 5G is fastest	Watching videos on the bus with no Wi-Fi
SIM / eSIM	Card or chip that identifies your phone to the carrier	Your phone's ID badge for the tower
GPS	Finds your location using satellites	The map app showing the blue dot where you are
MDM	Software a company uses to control work phones	A babysitter for all the company phones
BYOD	Bring Your Own Device: using your personal phone for work	Checking work email on your own phone
COPE	Corporate-Owned, Personally Enabled: a work phone you may use personally	A company phone you can also text friends on

Term	What it is	Example
Synchronization	Keeping data the same on all your devices	A contact added on your phone shows up on your laptop

Section 2: Networking

2.1 Ports and Protocols

Videos: Introduction to IP · Common Ports

Term	What it is	Example
IP	The addressing system that gets data to the right device	The street address on an envelope
TCP	Sends data and checks that every piece arrived	A letter with "please write back that you got it"
UDP	Sends data fast without checking	Shouting across the playground and hoping they heard
Port number	A numbered door for each service on a computer	The apartment number inside a building
HTTP (80) / HTTPS (443)	Loading websites; HTTPS is encrypted	The lock icon in your browser means HTTPS
SSH (22) / Telnet (23)	Remote control of a computer by typing; SSH is secure	Talking through a sealed tube vs. a paper cup on a string
FTP (20/21)	Moving files between computers	Passing a box of files across the room
SMTP (25)	Sends email	The mail carrier picking up your letter
POP3 (110) / IMAP (143)	Getting email; IMAP keeps it synced everywhere	IMAP lets you read the same email on phone and laptop
DNS (53)	Turns website names into IP addresses	The internet's phone book

Term	What it is	Example
DHCP (67/68)	Hands out IP addresses automatically	A teacher assigning seats when you walk in
SMB (445)	Windows file and printer sharing	Opening a shared folder on the office network
LDAP (389) / LDAPS (636)	Looking up users and accounts in a directory	Checking the school roster to find a student
RDP (3389)	Seeing and using another Windows computer's screen remotely	IT fixing your computer from their desk

2.2 Wireless Networks

Video: Wireless Network Technologies

Term	What it is	Example
802.11 (Wi-Fi)	The standard for wireless networks	Your home Wi-Fi
2.4 GHz vs 5 GHz	2.4 reaches farther; 5 is faster but shorter range	A deep voice carries through walls; a high voice is clearer up close
Channels	Lanes Wi-Fi uses so neighbors don't collide	Neighbors talking on different walkie-talkie channels
RFID	Tags read by radio waves	Tapping your badge to open a door
NFC	Very short-range wireless	Tap-to-pay with your phone

2.3 Network Services

Video: Network Services

Term	What it is	Example
File server	A computer that stores shared files	A shared class folder everyone can open

Term	What it is	Example
Print server	Manages printers for many people	One printer the whole office sends jobs to
Mail server	Sends, receives and stores email	The post office for your email
Syslog	Collects logs from many devices in one place	One diary that every device writes in
Web server	Hosts websites	The computer that gives you a webpage when you visit
Authentication server	Checks who you are before letting you in	A guard checking your ID
Spam gateway	Blocks junk email	A filter that tosses junk mail before your mailbox
All-in-one security appliance	One box doing firewall, antivirus and filtering	A Swiss Army knife for network security
Load balancer	Spreads traffic across many servers	Opening more checkout lanes when the store is busy
Proxy server	Goes to the internet on your behalf	A friend who goes into the store for you
SCADA / legacy / embedded / IoT	Industrial controls, old systems, built-in computers and smart devices	A smart thermostat or a factory's control panel

2.4 Network Configurations

Videos: DNS Configuration · DHCP · VLANs and VPNs

Term	What it is	Example
A / AAAA record	DNS entry linking a name to an IPv4 / IPv6 address	A phone book line: name → number
CNAME record	A nickname that points to another name	"Mom" in your contacts pointing to her real name

Term	What it is	Example
MX record	Tells email where to be delivered	The mailbox address for a company's email
TXT record	Extra text info in DNS, used for email security	A note pinned to the phone book entry
SPF / DKIM / DMARC	Email checks that stop fake senders	A signature and ID check on every letter
DHCP scope / pool	The range of addresses DHCP can hand out	The list of open seats in a classroom
DHCP reservation	One device always gets the same address	A reserved seat with your name on it
VLAN	Splits one network into separate groups	Invisible walls dividing one classroom
VPN	An encrypted tunnel over the internet	A secret tunnel under the playground

2.5 Network Devices

Video: Network Devices

Term	What it is	Example
Router	Connects different networks, like home to the internet	A mail carrier who knows which town each letter goes to
Switch (managed / unmanaged)	Connects devices in one network; managed ones can be configured	A helper handing each note to the right kid
Access point	Adds Wi-Fi to a wired network	A Wi-Fi "speaker" on the ceiling
Patch panel	A wall of ports where room cables meet	A neat switchboard for all the building's cables
Firewall	Allows or blocks traffic based on rules	A bouncer with a guest list

Term	What it is	Example
PoE	Power and data through one network cable	A security camera powered by its network cable
Cable / DSL modem	Connects you to your internet provider	The box your internet company gave you
ONT	The box that connects fiber internet to your home	The fiber version of a modem
NIC	The part that connects a computer to a network	The computer's network plug

2.6 Network Configurations (IP addressing)

Videos: IPv4 and IPv6 · Assigning IP Addresses

Term	What it is	Example
IPv4	Four numbers like 192.168.1.10	An older-style house number
IPv6	Longer addresses made because IPv4 ran out	Adding more digits to phone numbers so everyone gets one
Public vs private IP	Public is seen by the internet; private is only inside your network	Your home's street address vs. your bedroom number
Subnet mask	Shows which part of the address is the network	The "town" part of an address
Default gateway	The router address traffic uses to leave the network	The front door of the house
Static vs dynamic	Typed in by hand vs. given by DHCP	Your own name tag vs. one handed to you
APIPA (169.254.x.x)	Address Windows gives itself when DHCP fails	Making your own name tag because nobody gave you one

2.7 Network Connections

Videos: Internet Connection Types · Network Types

Term	What it is	Example
Fiber	Internet over light in glass; very fast	The fastest home internet plans
Cable	Internet over the TV coax line	Internet from your cable TV company
DSL	Internet over a phone line	Older internet sharing the home phone wire
Satellite	Internet from space; slow to respond	Internet at a remote cabin
Cellular	Internet from phone towers	A 5G home internet box
WISP	Wireless internet provider using antennas	An antenna on a farmhouse roof
PAN / LAN / WLAN	Personal, local, and wireless local networks	Earbuds (PAN), your house (LAN), home Wi-Fi (WLAN)
WAN	Networks spread over big distances	The internet is the biggest WAN
SAN	A network just for storage	A giant shared hard drive for servers

2.8 Network Tools

Video: Network Tools

Term	What it is	Example
Crimper	Attaches plugs to cable ends	A stapler that clamps the plug onto the wire
Wi-Fi analyzer	Shows nearby networks and signal strength	An app showing which Wi-Fi is strongest
Tone generator (toner)	Finds a cable by making it beep	Marco Polo with wires
Punchdown tool	Pushes wires into a wall jack or panel	Pressing each wire into its slot

Term	What it is	Example
Cable tester	Checks a cable's wires are connected right	A doctor's checkup for a cable
Loopback plug	Tests a port by sending its signal back to itself	Talking into a mirror to check your voice works
Network tap	Copies traffic so it can be watched	A listening device on a phone line

Section 3: Hardware

3.1 Displays

Videos: Display Types · Display Attributes

Term	What it is	Example
LCD (IPS, TN, VA)	Screen that needs a backlight; IPS has the best colors	Most laptop and office monitors
OLED	Each pixel makes its own light, so blacks are truly black	Every dot is its own tiny flashlight
Mini LED	LCD with thousands of tiny backlights	A brighter, sharper TV
Digitizer	The see-through touch layer on a screen	The part that feels where your finger pokes
Inverter	Powered the backlight in old screens	An old part you'll only see in older laptops
Resolution / pixel density	How many dots make up the picture	More dots = sharper picture
Refresh rate	How many times per second the screen redraws (Hz)	A flip book with more pages per second looks smoother
Color gamut	How many colors a screen can show	A bigger crayon box

3.2 Cables and Connectors

Videos: Network Cables · 568A and 568B Colors · Optical Fiber · Peripheral Cables · Video Cables · Storage Cables · Adapters and Converters · Copper Connectors · Fiber Connectors

Term	What it is	Example
Twisted pair (UTP / STP)	Network cable with twisted wire pairs; STP is shielded	Braided strings stay quieter than loose ones
Plenum cable	Fire-resistant cable for ceiling air spaces	Cable that's safe to run above an office ceiling
Direct burial	Cable made to go underground	Running network cable to a backyard shed
T568A / T568B	The two color orders for wiring a network plug	A recipe for which colored wire goes where
Fiber (single-mode / multimode)	Data as light; single-mode goes much farther	A flashlight message through a glass straw
USB / USB-C	Common plugs for devices; USB-C fits either way	Charging and connecting almost everything
Thunderbolt	Very fast data, video and power over a USB-C shape	One cable running two monitors
Serial / DB-9	Old 9-pin connection, sends one bit at a time	Configuring an old network switch
HDMI / DisplayPort	Modern video cables; HDMI carries sound too	Connecting a game console to a TV
DVI / VGA	Older video cables; VGA is the blue analog plug	An old school projector cable
SATA / eSATA	Cables for drives inside / outside the computer	A straw between the drive and the motherboard
Adapter / converter	Changes one plug type into another	A travel plug adapter for another country
RJ45 / RJ11	Network plug / smaller phone plug	The clicky plug on an Ethernet cable

Term	What it is	Example
F-connector	Screw-on plug for coax cable	The cable that screws into your TV box
Molex	Old 4-pin power plug	Powering an old drive or case fan
SC / ST / LC	Fiber plugs: square, round twist-lock, small	Different shaped keys for fiber locks

3.3 Memory

Videos: An Overview of Memory · Memory Technologies

Term	What it is	Example
RAM	Short-term working memory; cleared when power is off	Your desk; wiped clean at night
DIMM / SO-DIMM	Desktop memory stick / smaller laptop stick	A big desk for a big room, a small one for a small room
DDR3 / DDR4 / DDR5	Memory generations; each fits only its own slot	Different-shaped puzzle pieces
Parity / ECC	Memory that detects (parity) or fixes (ECC) errors	A spell-checker for memory
Multi-channel	Matched sticks working together for more speed	Two lanes on a road instead of one

3.4 Storage Devices

Videos: Storage Devices · RAID

Term	What it is	Example
Hard drive (HDD)	Spinning disk storage	A record player finding a song
SSD	Storage with no moving parts; much faster	A notebook that opens to any page instantly

Term	What it is	Example
NVMe / PCIe	A very fast way for SSDs to connect	An express lane for storage
M.2 / mSATA	Small card-shaped drives	A drive the size of a stick of gum
SAS	Fast drive connection used in servers	Storage built for business computers
Flash memory	USB sticks and memory cards	A camera's SD card
Optical drive	Reads CDs, DVDs and Blu-rays	Playing a movie disc
RAID 0	Splits data across drives for speed; no backup	Two people writing half a story each; lose one, lose the story
RAID 1	Exact copy on two drives	Writing your homework twice
RAID 5 / 6	Spreads data plus backup info; survives 1 / 2 failed drives	A team that can still finish if one member is sick
RAID 1+0	Mirroring plus striping; fast and safe	Two teams, each with a backup partner

3.5 Motherboards

Videos: Motherboard Form Factors · Motherboard Expansion Slots · Motherboard Connections · Motherboard Compatibility · The BIOS · BIOS Settings · HSM and TPM · CPU Features · Expansion Cards · Cooling

Term	What it is	Example
Motherboard	The main board everything plugs into	The streets connecting every building in a city
ATX / Micro-ATX / ITX	Motherboard sizes: standard, smaller, tiny	Big, medium and small pizza boxes
PCI / PCIe slots	Slots for add-on cards; PCIe is the modern one	Extra outlets for new gadgets
Headers	Board pins for front USB, audio, buttons and fans	Plugging the case's power button into the board

Term	What it is	Example
Intel vs AMD	The two main CPU makers; boards are made for one or the other	A charger that only fits one brand of phone
BIOS / UEFI	Startup program that runs before Windows	An alarm clock that wakes the computer
BIOS settings	Boot order, passwords, Secure Boot and more	Choosing which drive the computer starts from
TPM	A chip that safely stores encryption keys	A tiny safe built into the computer
HSM	A separate device that protects many keys for servers	A bank vault for keys
CPU	The brain that does the thinking	The chef cooking every order
32-bit vs 64-bit	64-bit can handle much more memory	A bigger plate holds more food
ARM	Power-saving processor design	What runs most phones
Cores	Separate workers inside the CPU	More chefs in the kitchen
Expansion cards	Add-on cards: sound, video, capture, network	Adding a better graphics card for games
Fans / heat sink / thermal paste	Parts that pull heat away from the CPU	Blowing on hot soup
Liquid cooling	Uses liquid to carry heat away	A car's radiator, but for a computer

3.6 Power

Video: Computer Power

Term	What it is	Example
Power supply (PSU)	Turns wall power into power the computer can use	A translator for electricity

Term	What it is	Example
Wattage	How much power the PSU can give	A bigger water tank for more thirsty parts
Redundant PSU	A backup power supply	A spare tire
Efficiency rating (80 PLUS)	How little power is wasted as heat	A car that gets better gas mileage

3.7 Multifunction Devices

Video: Multifunction Devices

Term	What it is	Example
Multifunction device	A printer that also scans, copies and faxes	The office machine that does everything
Printer language (PCL / PostScript)	How the computer describes the page to the printer	Giving the printer drawing instructions
Badging / secure print	Your job prints only after you badge in or enter a PIN	Your printout waits until you show up
Flatbed scanner	Glass you lay pages on to scan	A photocopier lid you lift up

3.8 Printers

Videos: Laser Printer Maintenance · Inkjet Printers · Inkjet Printer Maintenance · Thermal Printers · Thermal Printer Maintenance · Impact Printers · Impact Printer Maintenance

Term	What it is	Example
Laser printer	Uses toner powder and heat	Draw with static, sprinkle powder, iron it on
Maintenance kit	Replacement rollers and fuser for a laser printer	A tune-up kit for a car
Inkjet printer	Sprays tiny drops of ink	A robot flicking paint dots

Term	What it is	Example
Print head / carriage	The part that sprays ink and the track it slides on	A paint sprayer on a rail
Calibration	Lining up the printer so prints look right	Straightening a crooked picture frame
Thermal printer	Heats special paper to print	Store receipts
Heating element	The hot strip in a thermal printer	A tiny iron that darkens the paper
Impact (dot-matrix) printer	Pins hit an ink ribbon	Old multi-copy forms at a repair shop
Tractor feed	Paper with holes on the sides pulled by sprockets	Like film moving through an old projector

Section 4: Virtualization and Cloud Computing

4.1 Virtualization

Videos: Virtualization Concepts · Virtualization Services

Term	What it is	Example
Virtual machine (VM)	A pretend computer running inside a real one	A dollhouse inside your house
Sandbox	A safe, walled-off place to test things	A sandbox where nothing you do leaves the box
Legacy software	Old programs that need an old operating system	Running an old game in a VM of an older Windows
Hypervisor	Software that creates and runs VMs	The landlord sharing out rooms to each dollhouse
Type 1 hypervisor	Runs directly on the hardware	Built into the foundation of the house
Type 2 hypervisor	Runs as an app on a normal OS	VirtualBox on your laptop

Term	What it is	Example
VDI	Your desktop runs on a server; you connect to it	Your whole computer lives in another building
Container	A light package with just an app and what it needs	Lunchboxes in one fridge instead of a kitchen per meal

4.2 Cloud Computing

Videos: Cloud Models · Cloud Characteristics

Term	What it is	Example
Public / private cloud	Shared provider vs. one organization's own cloud	Renting a public gym vs. owning a home gym
IaaS	Rent raw servers; you set up everything else	Renting an empty kitchen
PaaS	Rent a ready platform to build your own apps	A kitchen stocked with ingredients; you cook
SaaS	Use finished software in a browser	Ordering pizza; someone else does it all
Internal vs external cloud	Run inside your company vs. by an outside provider	Your own garden vs. buying at the market
Metered usage	Pay only for what you use	An electric bill
Elasticity	Grows or shrinks with demand automatically	A stretchy backpack
Availability	Stays running even if parts fail	A store that never closes
File synchronization	Files match across all your devices	Saving on your phone, opening on your laptop
Multitenancy	Many customers share one system privately	Locked apartments in one building

Section 5: Hardware and Network Troubleshooting

5.1 Troubleshooting Hardware

Video: Troubleshooting Hardware

Problem	What it means	Example
POST beeps	The startup self-check found a problem	The computer "coughing" to say RAM is loose
Blue screen (BSoD)	Windows crashed, usually a driver or memory issue	The computer fainting and showing an error code
Blank screen	Power, cable, RAM or video card problem	Lights on but nobody home
No power	PSU, cable, outlet or power button	A lamp that won't turn on: check the plug first
Sluggish performance	Too hot, not enough RAM, or a failing drive	A runner wearing a heavy backpack
Overheating	Not enough cooling; may shut off on its own	The computer taking a nap so it doesn't get hurt

5.2 Troubleshooting Storage

Video: Troubleshooting Storage Devices

Problem	What it means	Example
Grinding or clicking	A hard drive is failing; back up now	A car making a scary noise
Drive not recognized	Loose cable, wrong boot order or dead drive	The computer can't find its toy box
Data loss / corruption	Files missing or broken	Pages torn out of your notebook
RAID degraded	One drive in the group failed	A team playing one player short

Problem	What it means	Example
S.M.A.R.T.	A drive's built-in health report	A fitness tracker for your hard drive
IOPS	How many reads and writes a drive does per second	How many bites per minute

5.3 Troubleshooting Displays

Video: Troubleshooting Display Issues

Problem	What it means	Example
Wrong input source	Monitor is looking at the wrong port	A TV set to HDMI 2 when the console is on HDMI 1
Projector bulb dim or dead	The lamp is wearing out	An old flashlight getting weaker
Fuzzy image	Not using the screen's best resolution	Glasses with the wrong prescription
Burn-in	A ghost image stuck on the screen	A sticker shadow left on a window
Dead pixels	Dots stuck black or one color	One burnt-out bulb in a big light sign
Flashing screen	Loose cable or failing backlight	A flickering light in a hallway

5.4 Troubleshooting Mobile Devices

Video: Troubleshooting Mobile Devices

Problem	What it means	Example
Poor battery life / swollen battery	Old or damaged battery; swollen is dangerous	A puffy battery is a sick battery; don't poke it
Broken screen	Cracked glass or display	A cracked phone screen

Problem	What it means	Example
Not charging properly	Bad cable, charger, or dirty port	Lint stuck in the charging port
Liquid damage	Water got inside	A phone dropped in the pool
Overheating	Too many apps or a bad battery	A phone too hot to hold
Digitizer issues / cursor drift	Touch not working or the cursor moves on its own	A ghost tapping your screen

5.5 Troubleshooting Networks

Video: Troubleshooting Networks

Problem	What it means	Example
No connectivity	Cable, Wi-Fi, or IP settings problem	A phone with no bars
Intermittent wireless	Connection keeps dropping	A radio station fading in and out
Jitter	Data arrives at uneven speeds	A friend talking in bursts on a video call
Poor VoIP quality	Internet calls sound choppy	A robot-sounding voice on a call
Port flapping	A switch port keeps turning on and off, often a bad cable	A light switch flicking on and off
High latency	Long delay before data arrives	Waiting a long time for an echo

5.6 Troubleshooting Printers

Video: Troubleshooting Printers

Problem	What it means	Example
Bad output (lines, streaks, faded)	Dirty drum, clogged nozzles, or low toner/ink	A marker running dry

Problem	What it means	Example
Toner smears off	The fuser isn't melting the toner	A sticker that never got pressed down
Garbled prints	Wrong or broken printer driver	The printer speaking the wrong language
Paper jams	Worn rollers, wrong paper, or debris	Paper stuck in the copier at school
Stuck print queue	Jobs won't print; restart the print spooler	A line that won't move until you reset it
Grinding noise	Gears or rollers failing	A bike chain that needs fixing
Finishing issues	Stapling or hole-punching not working	The stapler on the printer is empty
Wrong page orientation	Portrait vs landscape setting is wrong	A picture printed sideways